Spinning a Semantic Web for Agriculture

CGIAR | Platform for Big Data in Agriculture

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Sr. Research Fellow, Big Data Module Lead

SWAT4LS (Antwerp), Dec 4, 2018





CGIAR – agricultural research for development

Research conducted by 15 non-profit research Centers with 10,000 scientists and support staff in over 70 countries.

Close collaborations with national and regional research institutes, civil society orgs, academia, development orgs and private sector.



CENTERS

1. AfricaRice
2. Bioversity International
3. Center for International Forestry Research (CIFOR)
4. International Center for Agricultural Research in the Dry Areas (ICARDA)
5. International Center for Tropical Agriculture (CIAT)
6. International Crops Research Institute for the Semi-Arid Tropics (ICRISAT)
7. International Food Policy Research Institute (IFPRI)
8. International Institute of Tropical Agriculture (IITA)
9. International Livestock Research Institute (ILRI)
10. International Maize and Wheat Improvement Center (CIMMYT)
11. International Potato Center (CIP)
12. International Rice Research Institute (IRRI)
13. International Water Management Institute (IWMI)
14. World Agroforestry Centre (ICRAF)
15. WorldFish

Ag R4D: Complex networks, systems, infrastructure…

Multidisciplinary (agronomy, breeding, socioeconomics, bioinformatics, data science)

Multi-scale (genetic/genomic to landscape)

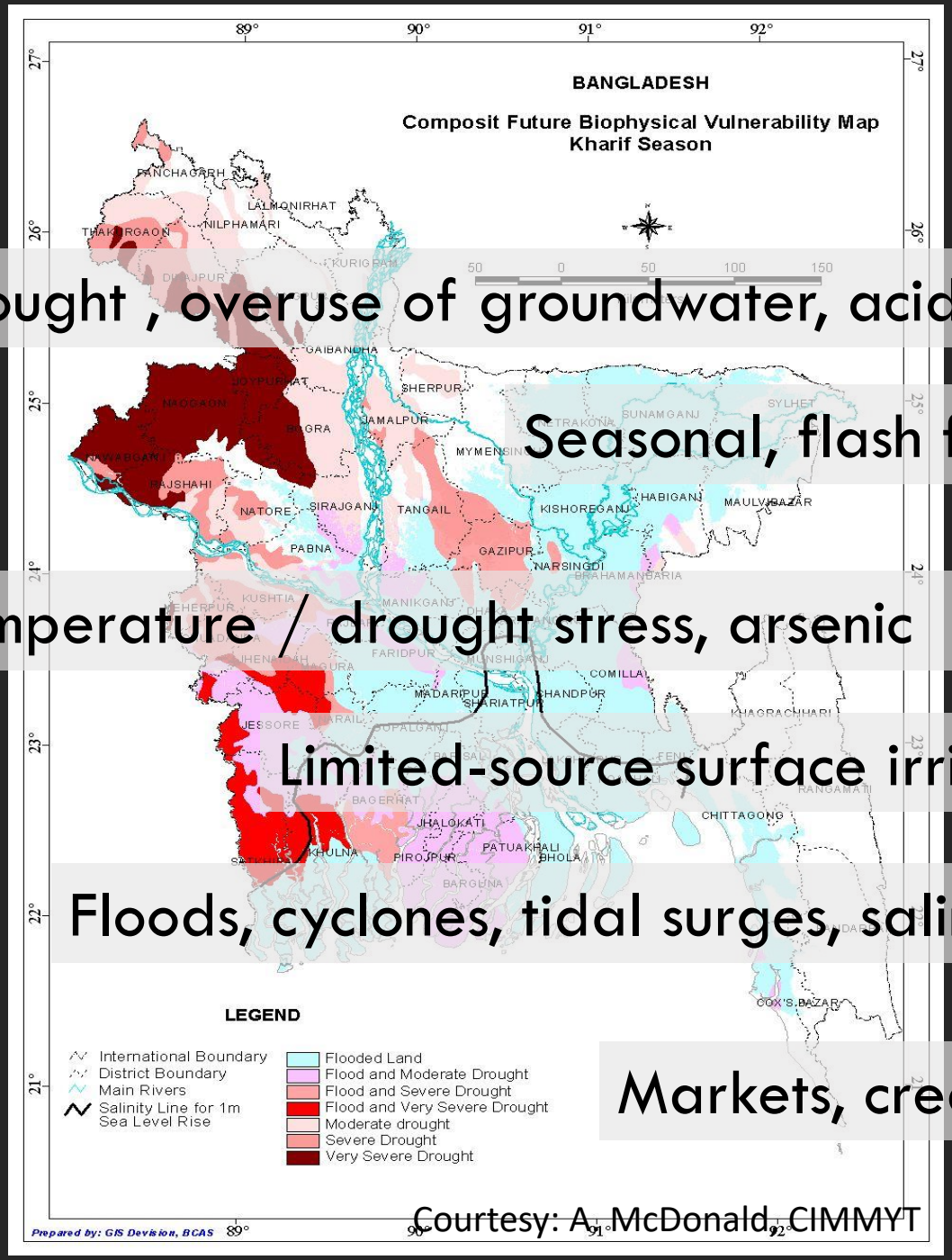
Multi-stakeholder consultative, participatory processes (planning to implementing)

Highly heterogeneous, climate-vulnerable, challenging environments









Drought , overuse of groundwater, acid soils

Seasonal, flash flooding

Temperature / drought stress, arsenic

Limited-source surface irrigation

Floods, cyclones, tidal surges, salinity

Markets, credit, insurance…

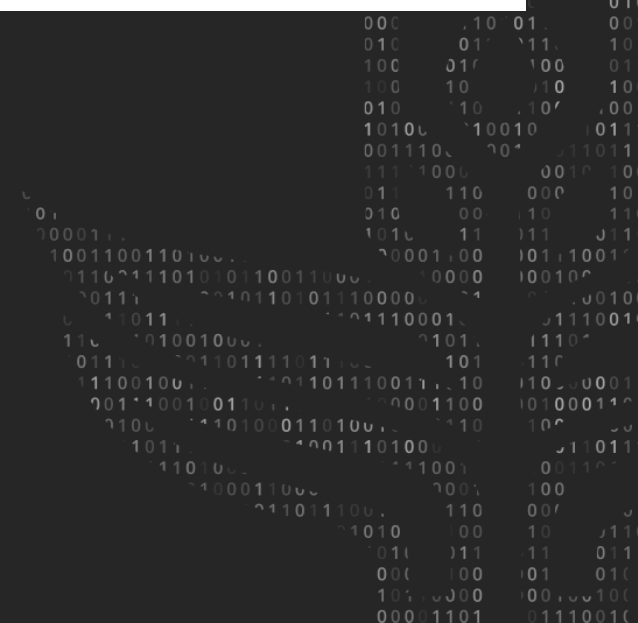
Courtesy: A. McDonald, CIMMYT

Hoffmann Centre for Sustainable Resource Economy

CHATHAM HOUSE
The Royal Institute of International Affairs

DeepMind Ethics & Society

Artificial Intelligence for a Sustainable and Healthy Food System





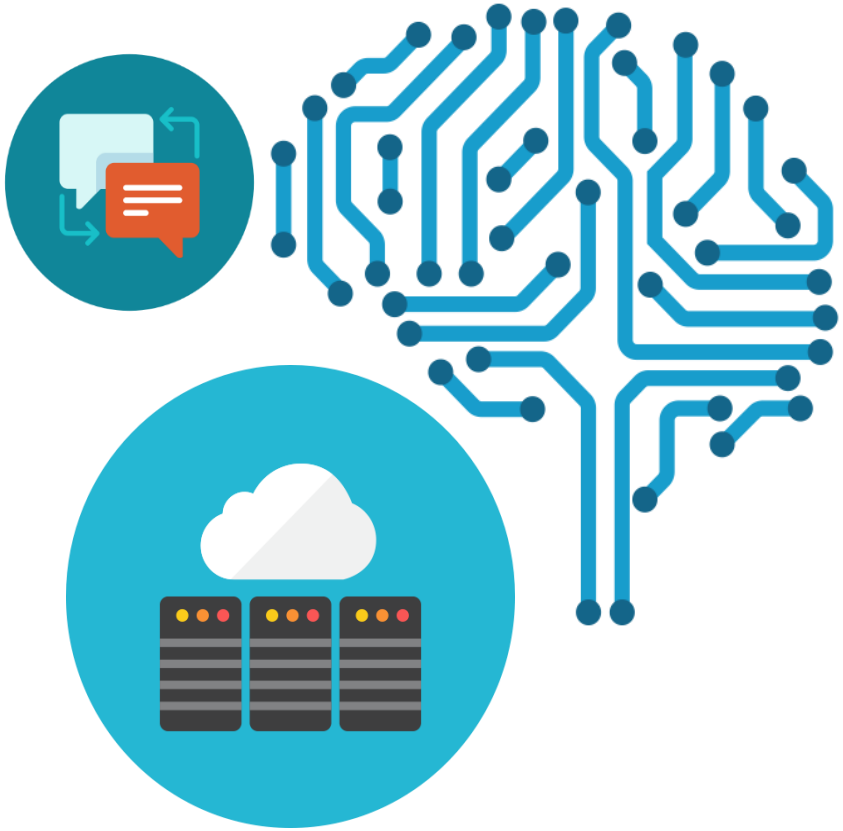
Hey Cigi, when should I plant my rice? How should I manage my crop?

Real-time **decision support, risk mitigation** for farmers

Easy **natural language** as an interface

Smart **artificial intelligence** trained by CGIAR and partners

Leveraging multiple open, **harmonized and interoperable** databases



Geotagged, time-stamped pictures of insured sites, sowing to harvest.

Timely ag advisories generated from the images + satellite imagery and localized data.



Images can be used in claims settlements.

Scientific innovation via Big Data approaches

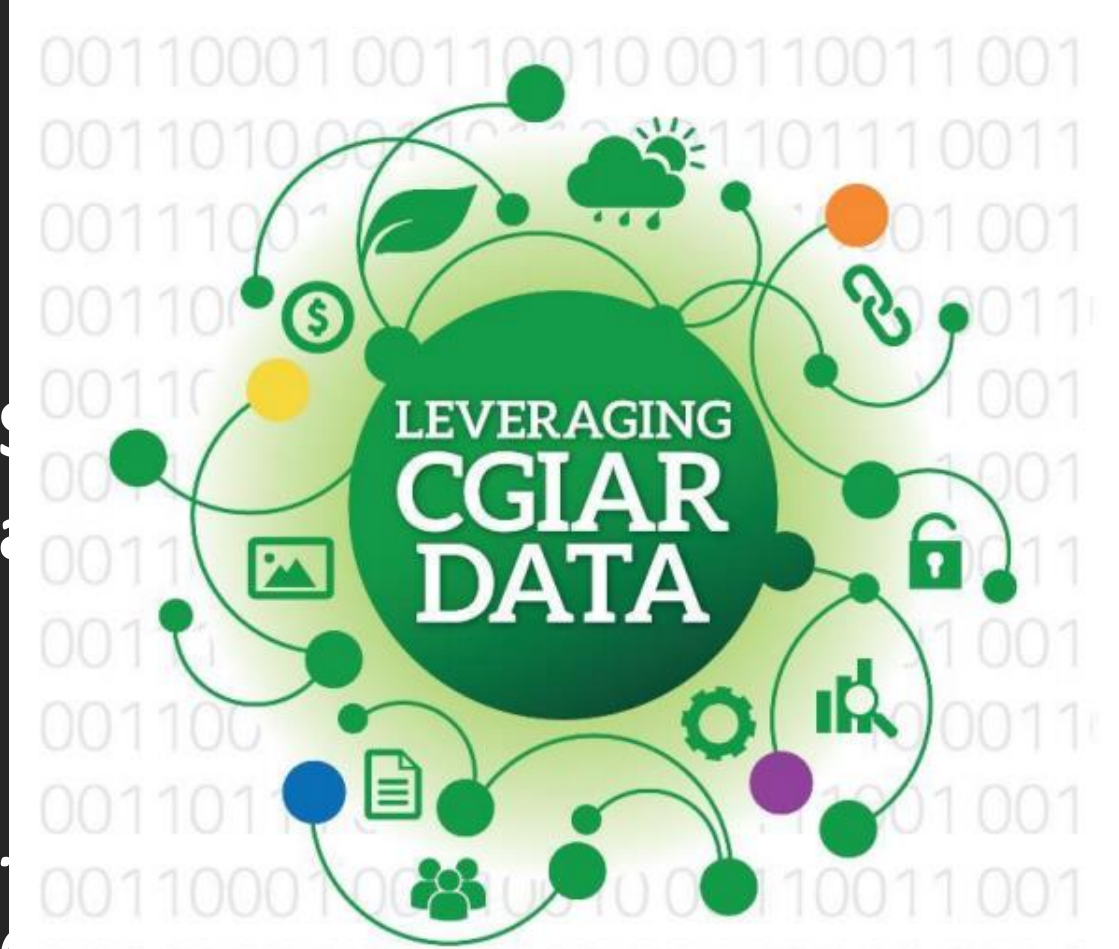


Find out what your soil says about your farm – and more!

- See how to best manage soil fertility
- Predict yield for different management and weather scenarios
- Get locations for trusted agro dealers near you

Opportunities



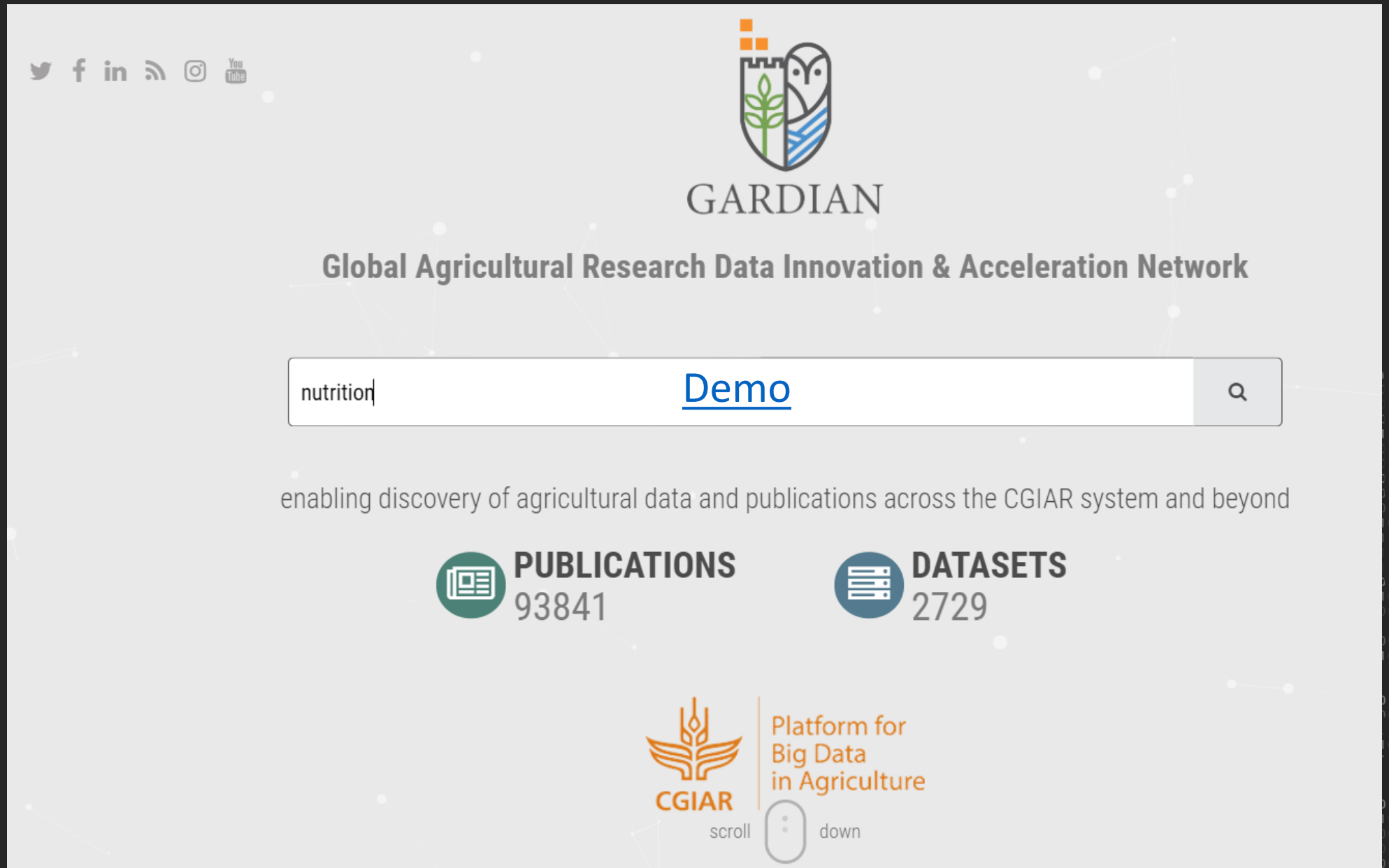




ting

ng,

Getting to FAIR… GARDIAN (http://gardian.bigdata.cgiar.org)



PUBLICATIONS (4408)

DATASETS (106)

GEOSPATIAL VIEW

2018 **Food Trees Nutrition Data Scooping Study in Uganda**

ICRAF

2018 **His and Hers, time and income: How intra-household dynamics impact nutrition in agricultural households**

CIAT

2018 **Probabilistic Causal Models for Nutrition Outcomes of Agricultural Actions - Kenya model**

ICRAF

2018 **Replication Data for: Urban Lusaka Food Consumption and Nutrition Survey: Role of Fish in Diets of Vulnerable groups**

WorldFish

2018 **Replication data for: Impact Evaluation of the Scaling Up Sweetpotato Through Agriculture and Nutrition (SUSTAIN) Project in Rwanda: Baseline Household Survey and Listing Data, 2014-2015**

CIP

2018 **Assessment of Informal Cross Border Fish Trade in the Southern African Region: A case of Zambia and Malawi**

WorldFish



VIEW LA

DATASET FILES

01. Household data collection Instrument.pdf
application/pdf

02. Individual data collection Instrument.pdf
application/pdf

g processes
on principle
nformation,

sion-making
consumption
in decision-

FAIR COMPL

F = 4.
A = 3.
I = 3.
R = 4.

VIEW ME

RELEVANT PUBLICATIONS

- What was the impact of dairy goats distributed by the Crop-Goat project in Tanzania?
- Poverty, household food security, and nutrition in rural Pakistan
- An integrated economic and social analysis to assess the impact of vegetable and fishpond technologies on poverty in rural Bangladesh
- Early childhood nutrition, schooling, and sibling inequality in a dynamic context, evidence

PUBLICATIONS (1992)

DATASETS (16)

GEOSPATIAL VIEW

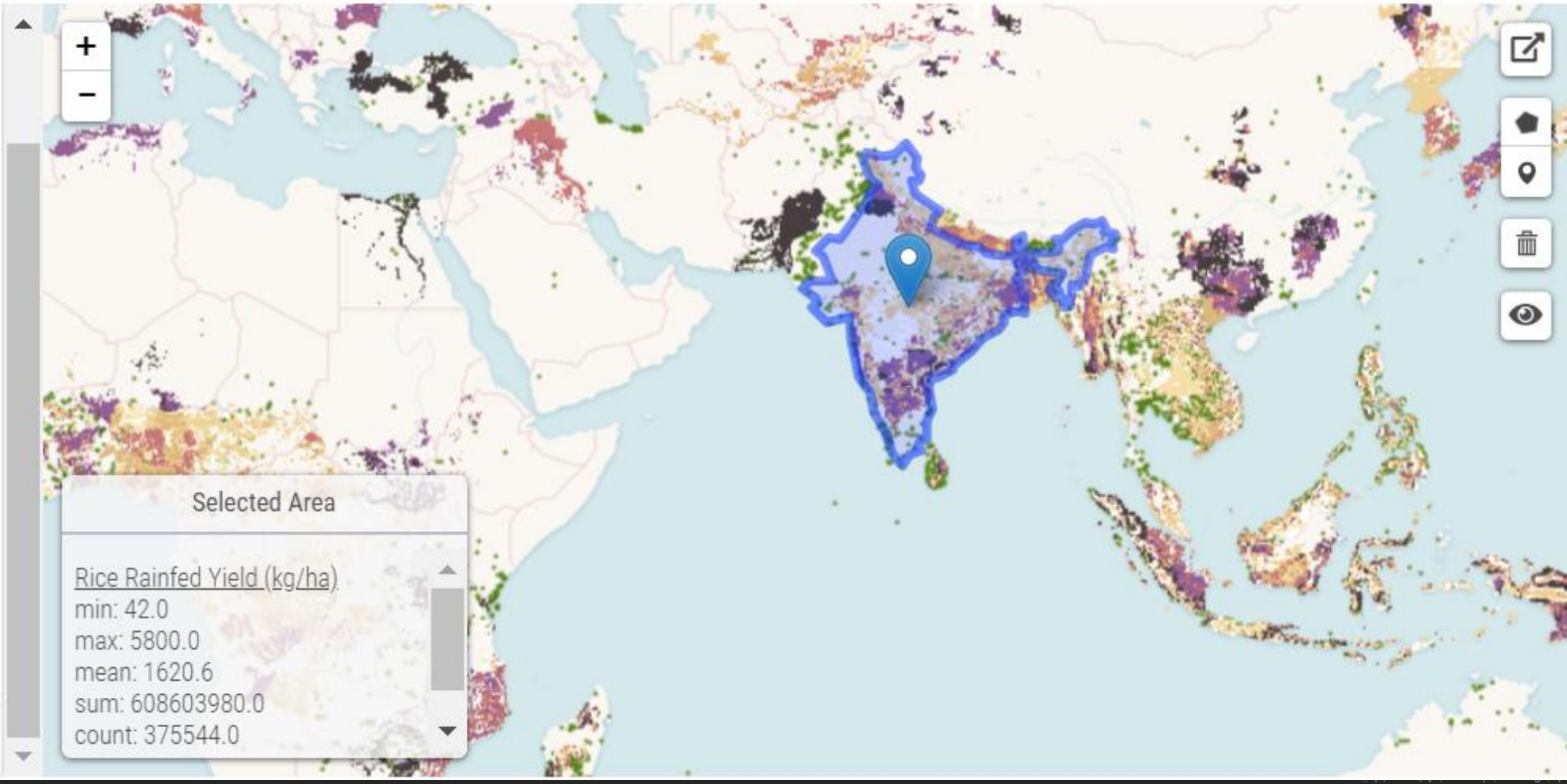
Rice

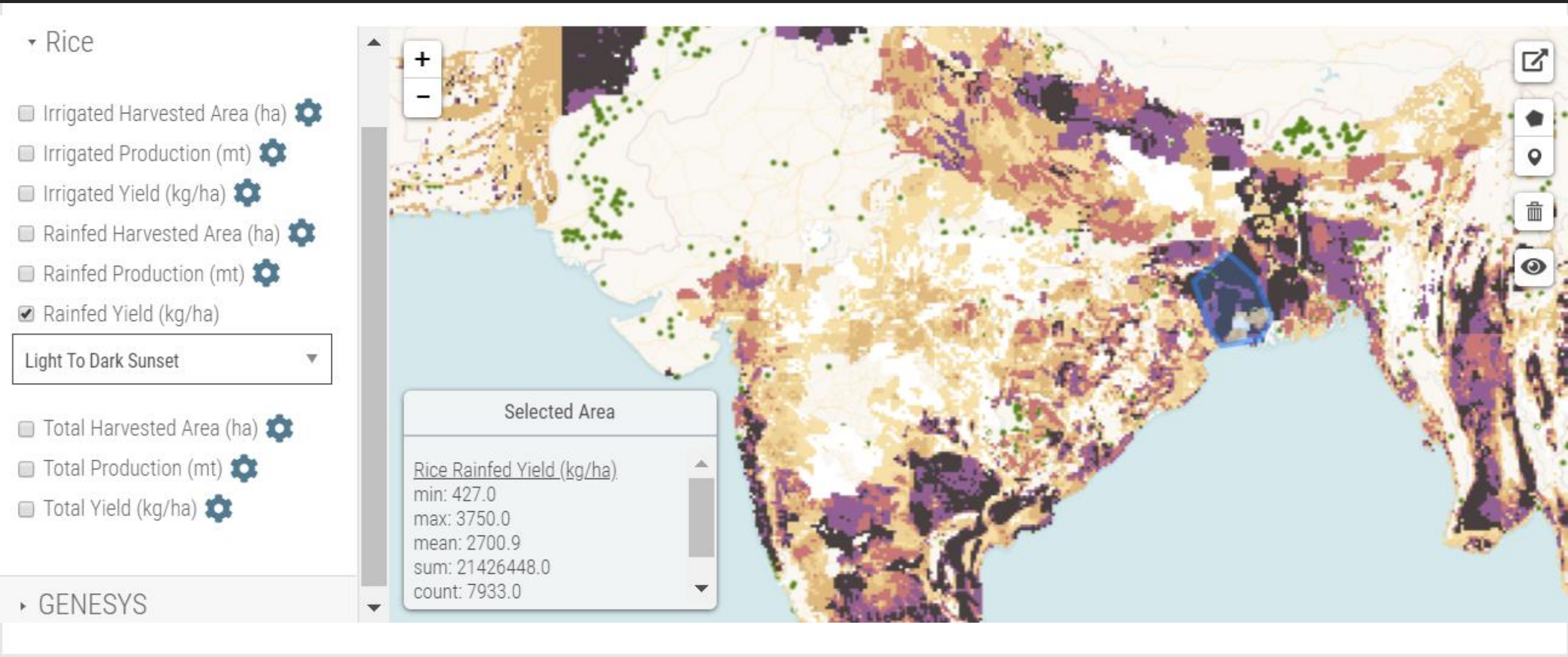
- [ ] Irrigated Harvested Area (ha)
- [ ] Irrigated Production (mt)
- [ ] Irrigated Yield (kg/ha)
- [ ] Rainfed Harvested Area (ha)
- [ ] Rainfed Production (mt)
- [x] Rainfed Yield (kg/ha)

Light To Dark Sunset

- [ ] Total Harvested Area (ha)
- [ ] Total Production (mt)
- [ ] Total Yield (kg/ha)

GENESYS



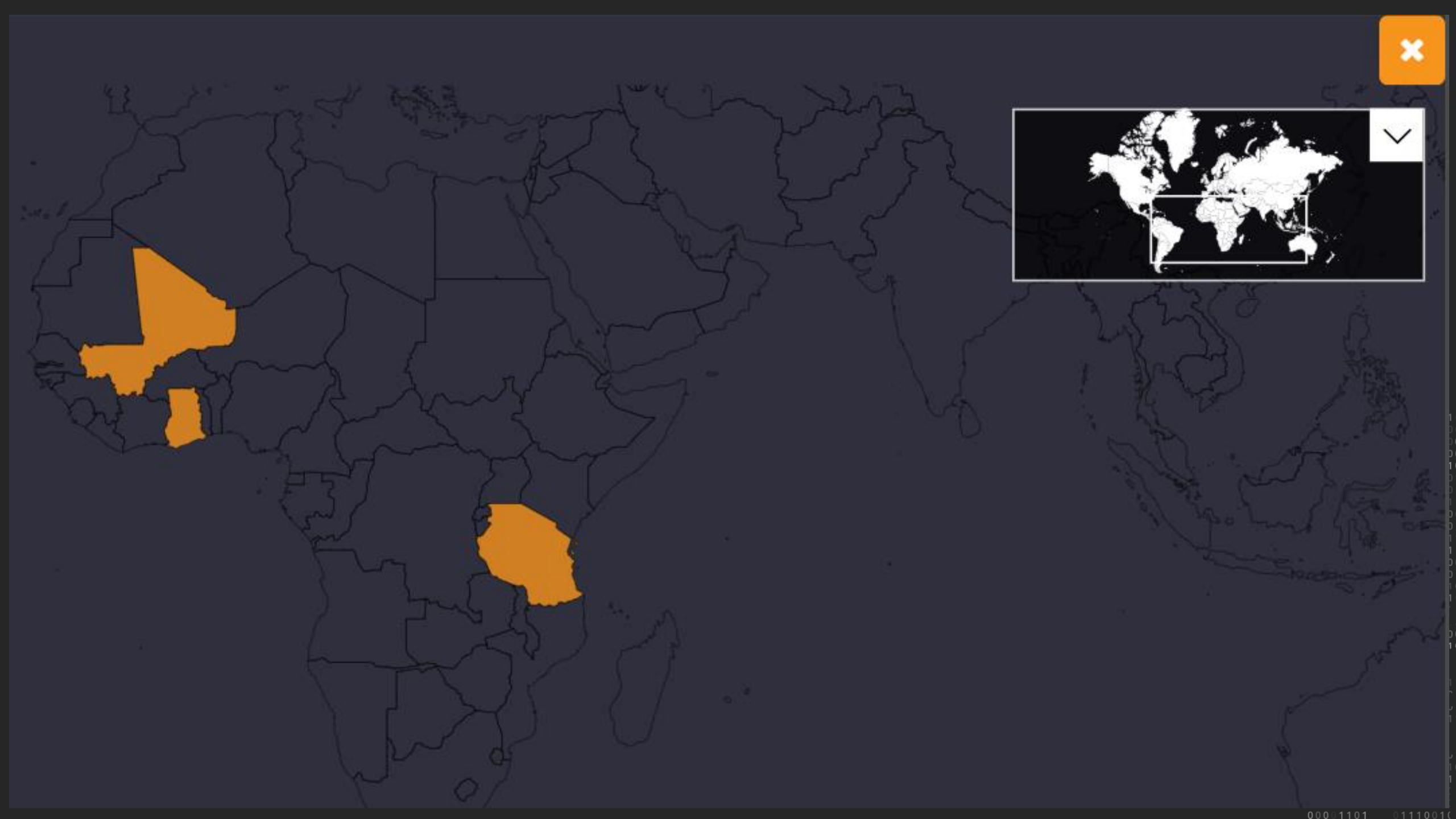

Rice
Irrigated Harvested Area (ha)
Irrigated Production (mt)
Irrigated Yield (kg/ha)
Rainfed Harvested Area (ha)
Rainfed Production (mt)
Rainfed Yield (kg/ha)
Light To Dark Sunset
Total Harvested Area (ha)
Total Production (mt)
Total Yield (kg/ha)
GENESYS
Selected Area
Rice Rainfed Yield (kg/ha)
min: 427.0
max: 3750.0
mean: 2700.9
sum: 21426448.0
count: 7933.0

Ask GARDIAN

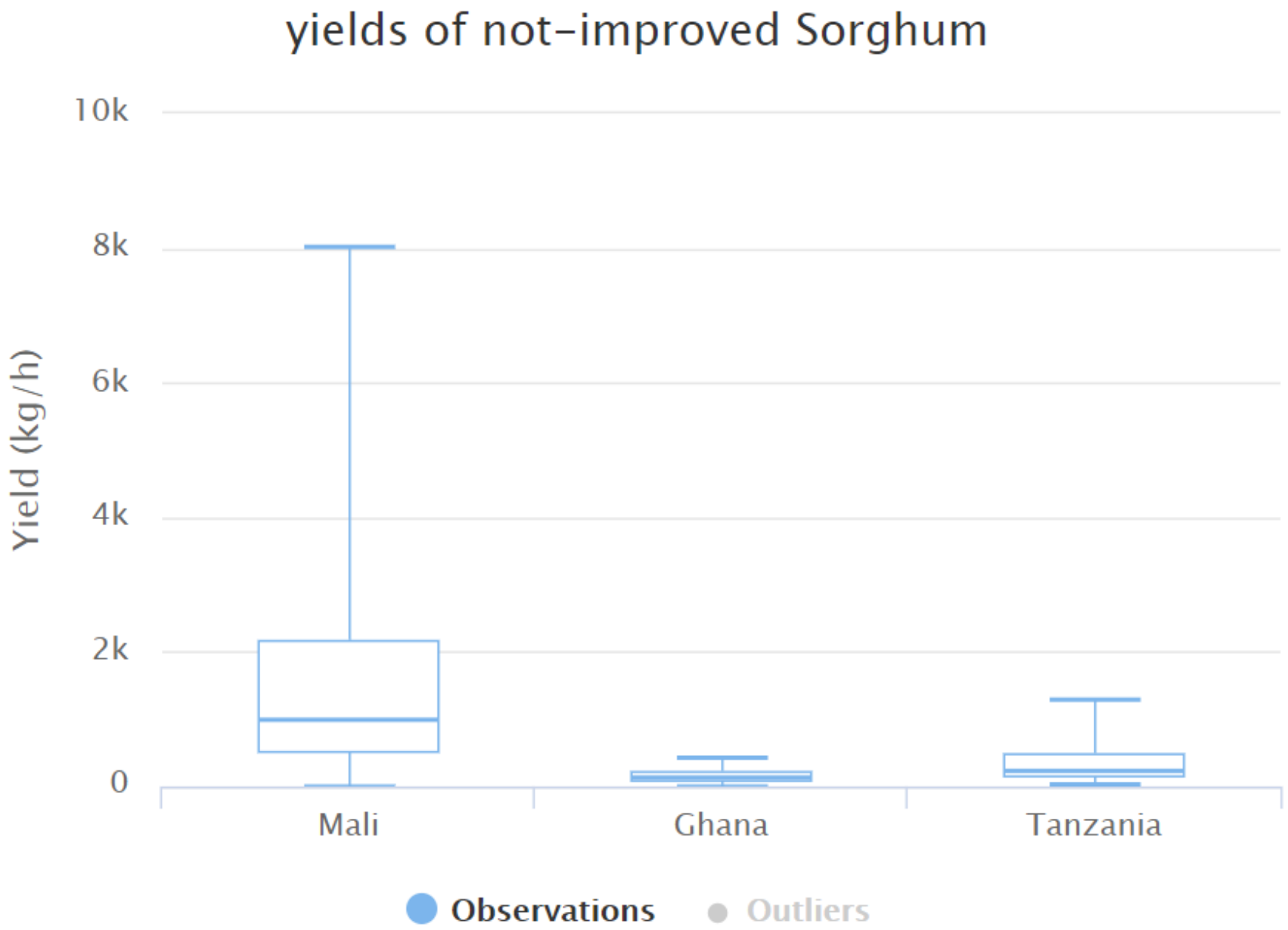
- Give me data for maize yields under *complete* and *no* fertilisation in **Africa**
- Give me data for yields of not-improved **maize** in **Africa**
- Give me data for maize yields in **Africa**
- Give me data for avg. maize yields of conservation vs. conventional practice in **Africa**
- Give me data for avg. maize yields of conservation vs. conventional tillage process in clay soil in **Africa**
- Give me data for **C content** in soil in **Tanzania**
- Give me data for soil texture of **topsoil** in **Tanzania**

Ask GARDIAN

- Give me data for maize yields under *complete* and *no* fertilisation in **Africa**
- Give me data for yields of not-improved **sorghum** in **Africa**
  - maize
  - rice
  - bambara
  - bean
  - groundnut
  - pearl millet
  - sorghum
- Give me data for maize yields in **Africa**
- Give me data for avg. maize yields of co conventional practice in **Africa**
- Give me data for avg. maize yields of conservation vs. conventional tillage process in clay soil in **Africa**
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- Give me data for soil texture of **topsoil** in **Tanzania**



SPARQL


yields of not-improved Sorghum
10k
8k
6k
4k
2k
0
Yield (kg/h)
Mali
Ghana
Tanzania
Observations
Outliers

Integrated Data

Get data as

Country	Village	Crop	Yield (kg/h)
Mali	Zansoni	Sorghum	60
Mali	Flola	Sorghum	150
Mali	Madina	Sorghum	343
Mali	Sirakele	Sorghum	8
Mali	Flola	Sorghum	750
Mali	N'Togonassa	Sorghum	0
Mali	Tiere	Sorghum	10
Mali	Goulala1	Sorghum	70
Mali	Tiere	Sorghum	0
Mali	Bobola-Zangasso	Sorghum	0

Previous 1 2 3 4 5 ... 67 Next

from Datasets

- Tanzania Africa Research in Sustainable Intensification for the Next Generation (Africa RISING) Baseline Evaluation Survey
- Ghana Africa Research in Sustainable Intensification for the Next Generation (Africa RISING) Baseline Evaluation Survey
- Mali Africa Research in Sustainable Intensification for the Next Generation (Africa RISING) Baseline Evaluation Survey

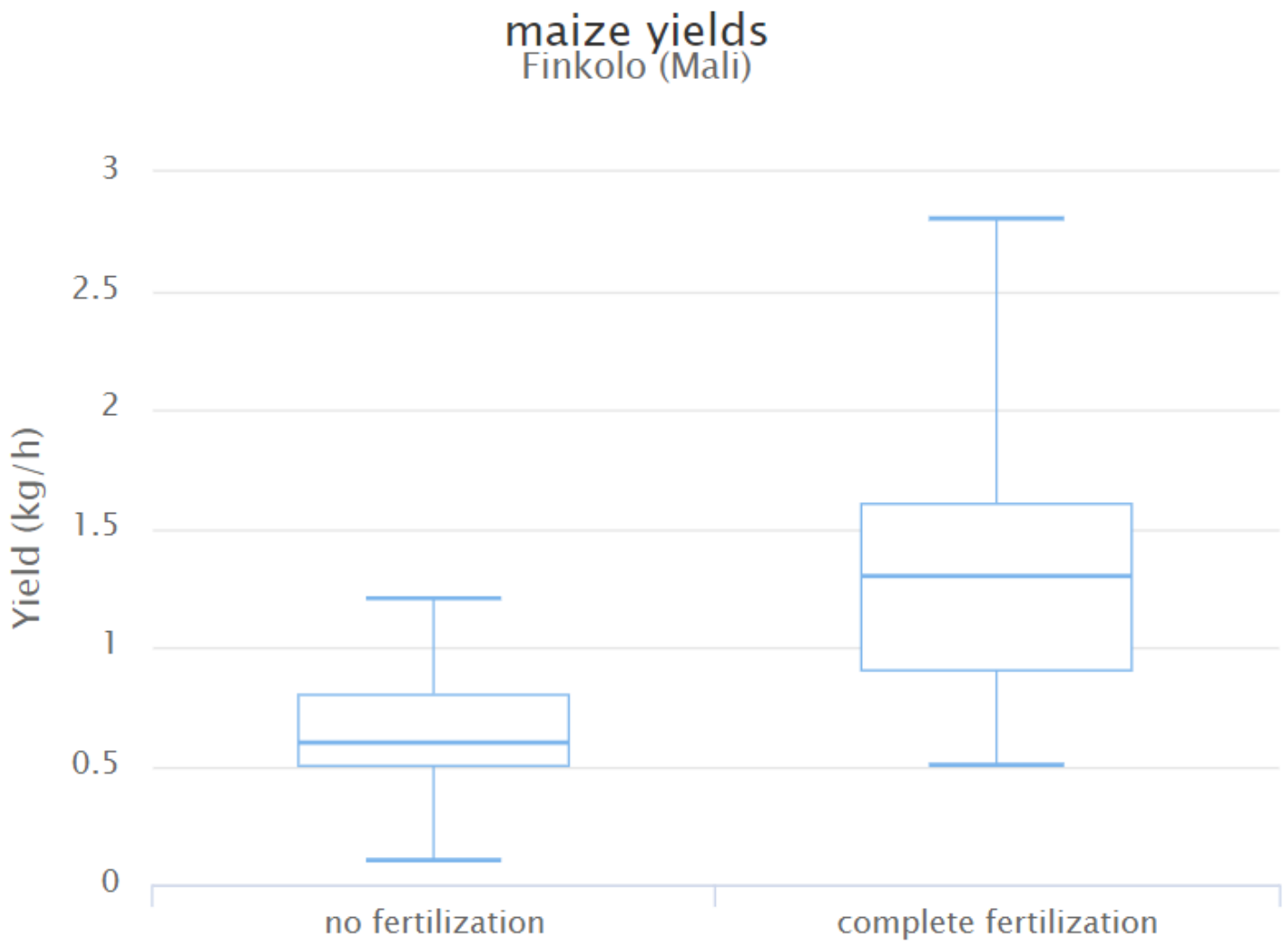
Ask GARDIAN

- Give me data for maize yields under *complete* and *no* fertilisation in **Finkolo (Mali)**
- Give me data for yields of not-improved **maize** in **Africa**
- Give me data for maize yields in **Africa**
- Give me data for avg. maize yields of conservation vs. conventional practice in **Africa**
- Give me data for avg. maize yields of conservation vs. conventional tillage process in clay soil in **Africa**
- Give me data for **C content** in soil in **Tanzania**
- Give me data for soil texture of **topsoil** in **Tanzania**



Results

SPARQL



Integrated Data

Get data as

Sample No	Region	Type	Yield (kg/h)
1	Finkolo (Mali)	Control	0.84375
2	Finkolo (Mali)	Control	0.125
3	Finkolo (Mali)	Control	1.0625
4	Finkolo (Mali)	Control	0.575
5	Finkolo (Mali)	Control	0.75
6	Finkolo (Mali)	Control	0.49375
7	Finkolo (Mali)	Control	0.25
8	Finkolo (Mali)	Control	0.48125
9	Finkolo (Mali)	Control	0.53125
10	Finkolo (Mali)	Control	0.59375

Previous 1 2 3 4 5 Next

from Datasets

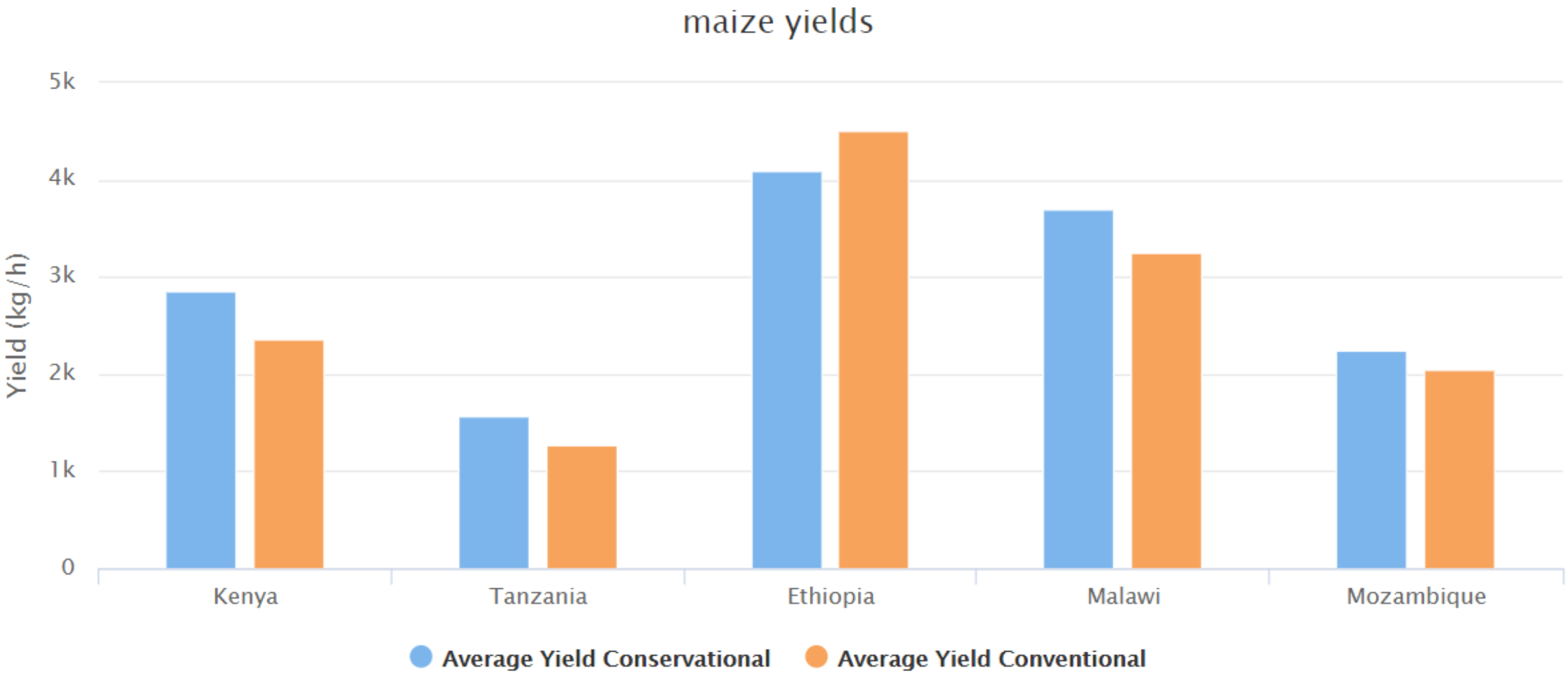
- Replication Data for: Maize response to macronutrients and potential for profitability in sub-Saharan Africa

Ask GARDIAN

- Give me data for maize yields under *complete* and *no* fertilisation in **Africa**
- Give me data for yields of not-improved **sorghum** in **Africa**
- Give me data for maize yields in **Africa**
- Give me data for avg. maize yields of conservation vs. conventional practice in **Kenya, Tanzania, Ethiopia, Malawi, Mozambique**
- Give me data for avg. maize yields of conservation vs. conventional tillage process in clay soil in **Africa**
- Give me data for **C content** in soil in **Tanzania**
- Give me data for soil texture of **topsoil** in **Tanzania**

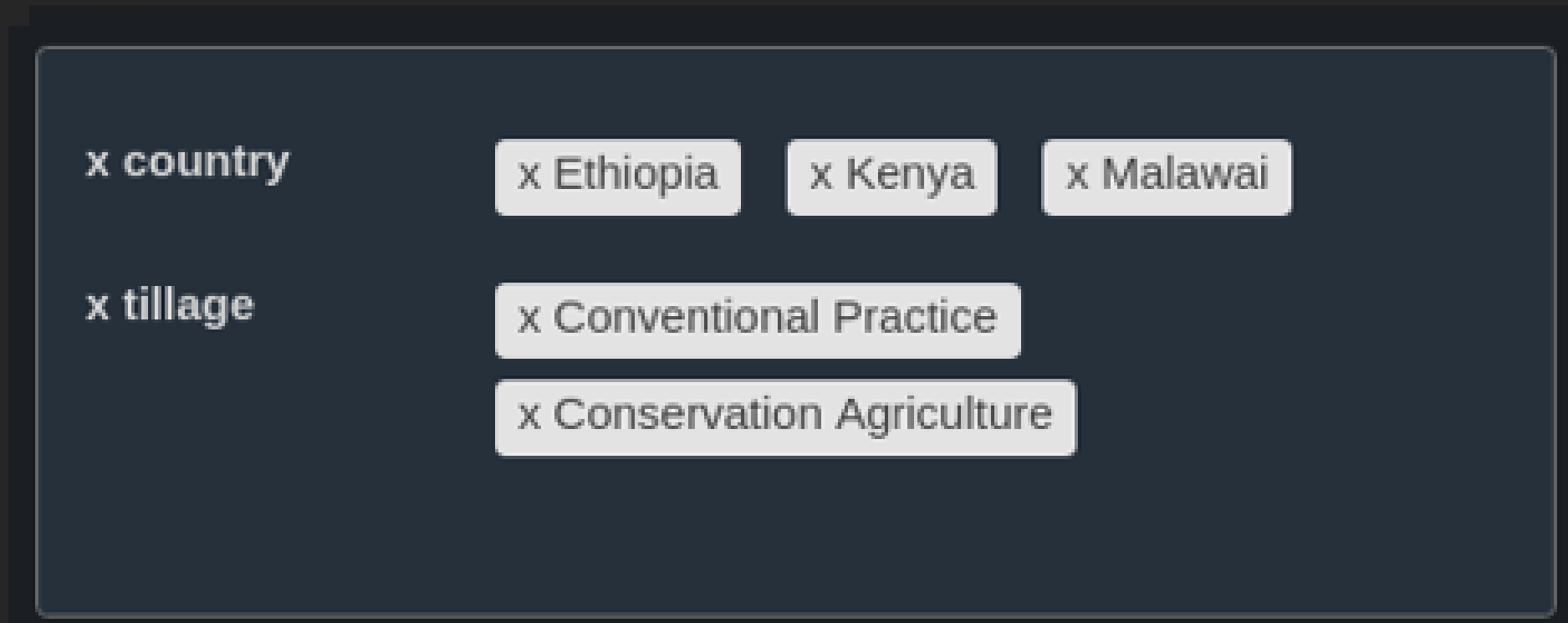
Results

SPARQL



What’s next?

1. Search and filter semantic data described with ontologies.



Ontocale

What's next?

2. View groupings of plots across projects matching criteria. See what other data or variables exist in each.

Common categories (all plots)

Basal fertilizer quantity (kg/ha)

Top dressing fertilizer N (kg/ha)

Total N (kg/ha)

Total P (kg/ha)

Country

Tillage

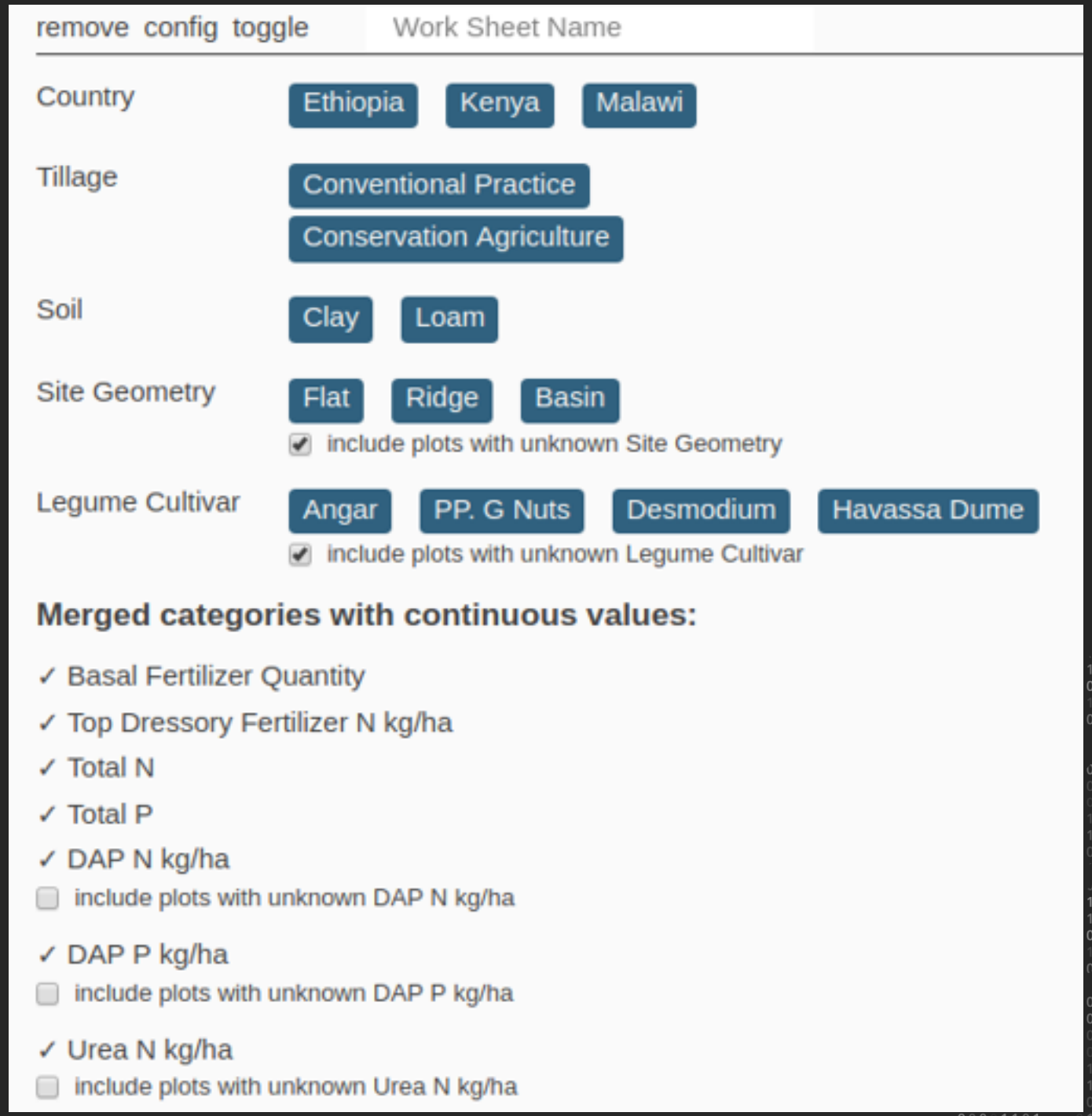
Soil

Sparse categories	527 plots	1062 plots	1026 plots
Site geometry	X	✓	✓
DAP N (kg/ha)	✓	✓	X
DAP P (kg/ha)	✓	✓	X
Urea N (kg/ha)	✓	X	X
Slope (%)	✓	✓	X
Legume cultivar	X	✓	X

What's next?

3. Select data to include in a merged dataset.

   Download in tabular format or aggregate for visual display.



Ontocale

	Variable	Description	Type	Unit	Average	Median	Minimum	Maximum	St Dev	No. missing values
SITE	Country name	Name of country in which experiments were conducted	Text	Text						
	SIMLESA Site name	Name of site eg district where experiments were	Text	Text						
	Farmercode	Farmner serial number	Numeric	Numeric						
	Treatment name/code	Refers to Cropping system used. CA= Conservation Agriculture; CP=conventional practice in the local country context	Text	Text						
	Legume association	Refers to legume intercropped or rotated with maize. INT=intercrop; SOLE= crop grown on its own; ROT=crops rotated	Text	Text						
	Geometry	How the field was prepared for planting. Options = flat, mounts, beds, ridges	Text	Text						
	Tillage Practice	Type of tillage system used. CP= conventional ploughing; R/F=conventional ridge and furrow; D/S=	Text	Text						
SITE DESCRIPTION	Season	Year of harvesting	Date	YYYY	2012.484	2012	2010	2016	1.74917354	1
	Slope	General estimate of slope of fields used for the trial	Numeric	%	3.200745	2.5	1	5.5	1.74076277	61
	Total rainfall	The total amount of rainfall received in that cropping season in mm	Numeric	mm	774.4452	729	213	1865.3	427.195465	178
	Rainfall in the first 30 days afte	Rainfall in the first 30 days after planting	Numeric	mm	215.0086	236	26	483	115.550559	208
	BD 0-20 cm	Soil bulk density in g/cm^3	Numeric	g/cm3	1.388691	1.4	1.16	1.48	0.04504259	408
	Textural class	Refers to broad soil classification in terms of fineness or coarseness of its texture in top 20 cm eg clay loam, clay, silty loam, sand	Text	Text						
	Sand 0-20 cm	% soil content by weight constituting the sand fraction	Numeric	%	29.65865	27	17	55.5	10.3819011	298
	Silt 0-20 cm	% soil content by weight constituting the silt fraction	Numeric	%	28.26323	19	19	59.26553	14.6012951	298
	Clay 0-20 cm	% soil content by weight constituting the clay fraction	Numeric	%	42.07643	54	12.5	54	16.632	298
TREATMENT DESCRIPTION	Maize cultivar	Maize variety used	Text	Text						
	Legume species	Type of legume crop used	Text	Text						
	Legume cultivar	Legume crop variety name	Text	Text						
	Surface mulch type 1	Material used as residue cover	Text	Text						
	Quantity surface mulch 1	rate of surface residue cover application	Numeric	kg/ha	0.528034	0.75	0	1	0.41476683	168
	Surface mulch type 2	In cases where mixed residue cover types are used, this indicates the second type of mulch used	Text	Text						
	Quantity surface mulch 2	Rate of surface residue cover type 2 application	Numeric	kg/ha						
	Basal fertilizer type	Name of basal fertilizer used	Text	Text						
	Basal fertilizer quantity	Rate of basal fertilizer application	Numeric	kg/ha	59.92057	100	0	125	52.3231753	78
		Phosphorus content of DAP basal fertilizer	Numeric	kg/ha	36.5838	46	0	69	24.2365779	226
	DAP (P&N)	Nitrogen content of Diammonium phosphate fertilizer	Numeric	kg/ha						
		Phosphorus content ofapplied NPK basal fertilizer	Numeric	kg/ha						
	NPK	Nitrogen content of applied NPK fertilizer	Numeric	kg/ha						
	Urea	Nitrogen content of applied Urea fertilizer	Numeric	kg/ha						

Mozamb	Cabango	159	3	Farmers check	SOLE	BASIN	CA	2013-14		984.9	382.4					
Mozamb	Cabango	157	4	Basins Maize sole	SOLE	FLAT	CP	2013-14		1024.3	346					
Mozamb	Cabango	157	4	jab planter +R/up+Maize sole	SOLE	BASIN	CA	2013-14		1024.3	346					
Mozamb	Cabango	157	4	jab planter Maize-common bean rota	ROTCP	FLAT	CA	2013-14		1024.3	346					
Mozamb	Cabango	157	4	jab planter Maize-common bean Inte	INTCP	BASIN	CA	2013-14		1024.3	346					
Mozamb	Cabango	157	4	Farmers check	SOLE	BASIN	CA	2013-14		1024.3	346					
Mozamb	Cabango	439	5	Basins Maize sole	SOLE	FLAT	CP	2013-14		618.3	85					
Mozamb	Cabango	439	5	jab planter +R/up+Maize sole	SOLE	BASIN	CA	2013-14		618.3	85					
Mozamb	Cabango	439	5	jab planter Maize-common bean rota	ROTCP	FLAT	CA	2013-14		618.3	85					
Mozamb	Cabango	439	5	jab planter Maize-common bean Inte	INTCP	BASIN	CA	2013-14		618.3	85					
Mozamb	Cabango	439	5	Farmers check	SOLE	BASIN	CA	2013-14		618.3	85					
Mozamb	Cabango	257	6	Basins Maize sole	SOLE	FLAT	CP	2013-14		930.7	67					
Mozamb	Cabango	257	6	jab planter +R/up+Maize sole	SOLE	BASIN	CA	2013-14		930.7	67					
Mozamb	Cabango	257	6	jab planter Maize-common bean rota	ROTCP	FLAT	CA	2013-14		930.7	67					
Mozamb	Cabango	257	6	jab planter Maize-common bean Inte	INTCP	BASIN	CA	2013-14		930.7	67					
Mozamb	Cabango	257	6	Testemunha do campones	SOLE	BASIN	CA	2013-14		930.7	67					
Mozamb	Sussundenga	165	1	Covacho Milho puro	SOLE	FLAT	CP	2013-14		1196	471.4					
Mozamb	Sussundenga	165	1	Matraca +R/up+Milho puro	SOLE	BASIN	CA	2013-14		1196	471.4					
Mozamb	Sussundenga	165	1	Matraca Feijao vulgar-Milho rotacac	ROTCP	FLAT	CA	2013-14		1196	100.8					
Mozamb	Sussundenga	165	1	Matraca Milho-Feijao vulgar consoc	INTCP	BASIN	CA	2013-14		1196	471.4					
Mozamb	Sussundenga	165	1	Testemunha do campones	SOLE	BASIN	CA	2013-14		1196	471.4					
Mozamb	Sussundenga	342	2	Covacho Milho puro	SOLE	FLAT	CP	2013-14		1493.6	471					
Mozamb	Sussundenga	342	2	Matraca +R/up+Milho puro	SOLE	BASIN	CA	2013-14		1493.6	471					

HIDAP AgroFIMS

Hello, Guest
Not connected
Log in

- Fieldbook
- Single Trial Analysis
- Documentation
- Help
- About

Platform for Big Data in Agriculture

HIDAP AgroFIMS

Agronomy Field Management System



HIDAP AgroFIMS v0.0.17

The Agronomy Field Information Management System (AgroFIMS) has been developed on CGIAR's HiDAP (Highly-interactive Data Analysis Platform created by CGIAR's International Potato Center, CIP). AgroFIMS draws fully on ontologies, particularly the Agronomy Ontology and the Crop Ontology. It consists of modules that represent the typical cycle of operations in agronomic trial management, and enables the creation of data collection sheets using the same ontology-based set of variables, terminology, units and protocols. AgroFIMS therefore:

- Standardizes data collection and description for easy aggregation and inter-linking across disparate datasets;
- Allows easy integration with HiDAP breeding data, or any other ontology-based datasets;
- Functions as a data staging repository, allowing data uploads with view/edit permissions;
- Enables data quality checks, statistical analysis of the data collected, and the generation of sophisticated statistics reports;
- Aligns a priori with CGIAR's CG Core metadata schema;
- Enables easy upload to the institutional repositories, and much more.

Funding for AgroFIMS was provided by the Bill and Melinda Gates Foundation's Open Access, Open Data Initiative, and the CGIAR Big Data Platform.

AgroFIMS: Key features

Standardized data collection based on ontologies (e.g. AgrO), methodologies

Built-in metadata (mapped to CGIAR repositories) = easy upload to repos

Built-in R scripts for statistical analysis with graphs, reports generated

Easier data integration = enhanced cross-regional, cross-disciplinary learning

Plug-n-play with Big Data platform's analytical, modeling, visualization tools

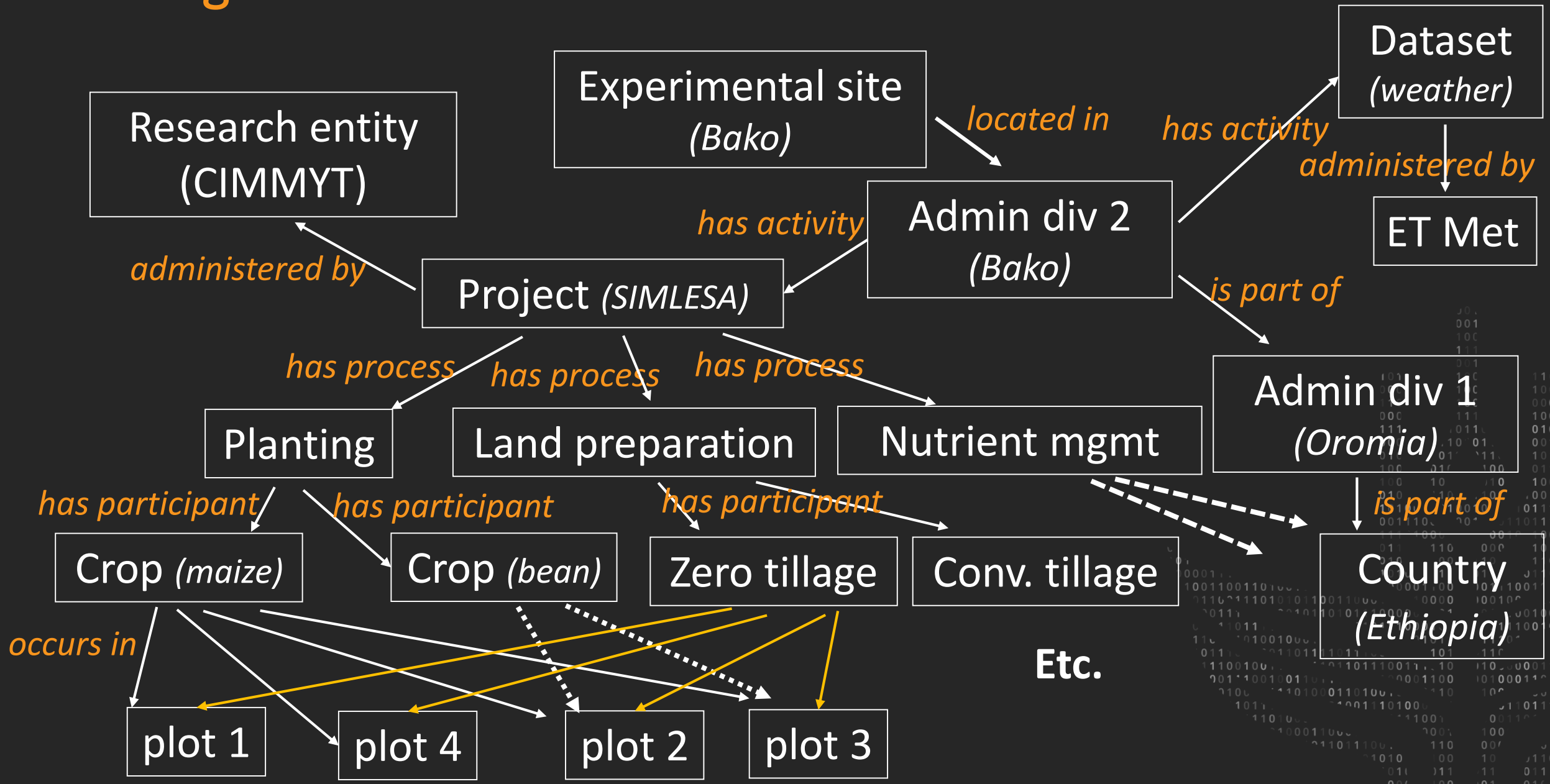
Ease of use (paper or mobile-based digital data collection)



	A	B	C	D	E	F	G	H	I
1	PLOT	BLOCK	TREATMENT	Irrigation_Number of irrig	Weeding_Weeding techni	Land preparation_Technic	Wheat-Timing-Date-yyyy/mm/dd	Wheat-Harvest-Area harvested-m	Wheat-Grain-Fresh weight-g
2	1	1	4_Chemical_No-till	4	Chemical	No-till			
3	2	1	4_Manual_No-till	4	Manual	No-till			
4	3	1	8_Manual_No-till	8	Manual	No-till			
5	4	1	4_Chemical_Conventional till	4	Chemical	Conventional till			
6	5	1	8_Chemical_Conventional till	8	Chemical	Conventional till			
7	6	1	4_Manual_Conventional till	4	Manual	Conventional till			
8	7	1	8_Manual_Conventional till	8	Manual	Conventional till			
9	8	1	8_Chemical_No-till	8	Chemical	No-till			
10	9	2	4_Chemical_Conventional till	4	Chemical	Conventional till			
11	10	2	4_Chemical_No-till	4	Chemical	No-till			
12	11	2	8_Manual_No-till	8	Manual	No-till			
13	12	2	4_Manual_No-till	4	Manual	No-till			
14	13	2	8_Chemical_Conventional till	8	Chemical	Conventional till			
15	14	2	8_Chemical_No-till	8	Chemical	No-till			
16	15	2	8_Manual_Conventional till	8	Manual	Conventional till			
17	16	2	4_Manual_Conventional till	4	Manual	Conventional till			
18	17	3	4_Manual_No-till	4	Manual	No-till			
19	18	3	8_Chemical_Conventional till	8	Chemical	Conventional till			
20	19	3	4_Chemical_Conventional till	4	Chemical	Conventional till			
21	20	3	8_Manual_Conventional till	8	Manual	Conventional till			
22	21	3	4_Manual_Conventional till	4	Manual	Conventional till			
23	22	3	8_Chemical_No-till	8	Chemical	No-till			
24	23	3	4_Chemical_No-till	4	Chemical	No-till			
25	24	3	8_Manual_No-till	8	Manual	No-till			

Metadata | Phenology | **Fieldbook** | Irrigation | Harvest | Weeding | Soil fertility | Trait list | Weather | Soil

What might this look like?



Search results for 'conservation agriculture' limited to type 'people'

Not the results you expected?

Limit people to

Person (35)

Singh, Ravi Gopal | Person

... Agent Continuant Entity Independent Continuant Person Philippines Ecology and management of weeds under **conservation agriculture**: A review ...

Christian, Thierfelder | Person

... Agent Continuant Entity Independent Continuant Person Mozambique Zimbabwe Zambia Malawi Facilitating the widespread adoption of **conservation agriculture** ...

Capis, Jude | Person

... Agent Continuant Entity Independent Continuant Person Towards an Evergreen **Agriculture** in Africa: Scaling-up **Conservation Agriculture** with Trees for improved ...

Kshatriya, Mrigesh | Person

... Agent Continuant Entity Independent Continuant Person Towards an Evergreen **Agriculture** in Africa: Scaling-up **Conservation Agriculture** with Trees for improved ...

Jonathan, Muriuki | Person

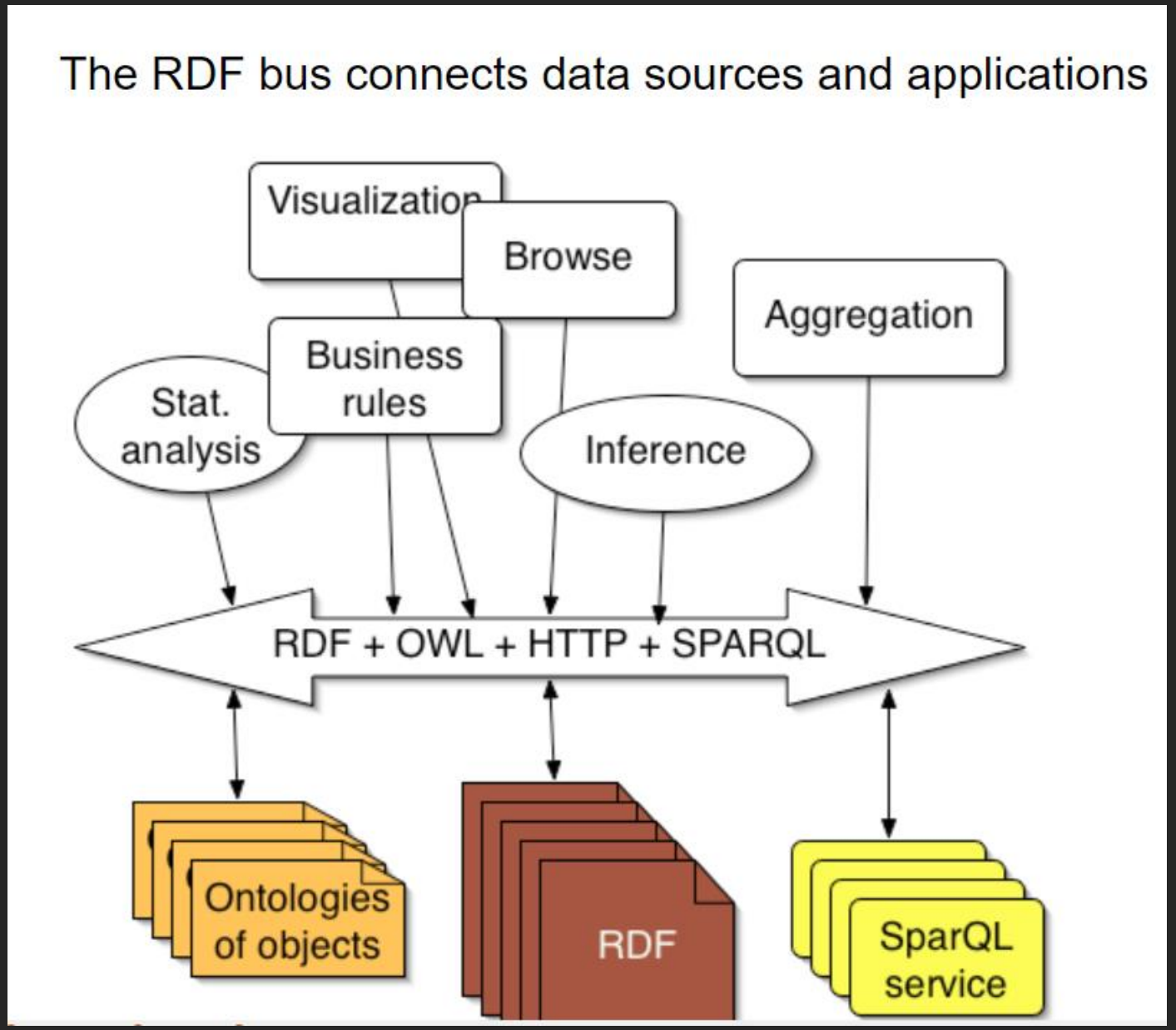
... Agent Continuant Entity Independent Continuant Person Evergreen **Agriculture** Baseline Survey CAWT Survey: Scaling-up the science and practice of **conservation** ...

Peter, Gachie | Person

... Agent Continuant Entity Independent Continuant Person CAWT Survey: Scaling-up the science and practice of **conservation agriculture** in sub-saharan Africa ...

and Environmental Resilience in Eastern and Southern Africa | Dataset

Get on the bus, Gus! Make a new plan, Stan – just listen to me…



https://www.w3.org/2006/Talks/0718-aaai-tbl/Overview.html#(2)

Hoffmann Centre for Sustainable Resource Economy

CHATHAM HOUSE
The Royal Institute of International Affairs

DeepMind Ethics & Society

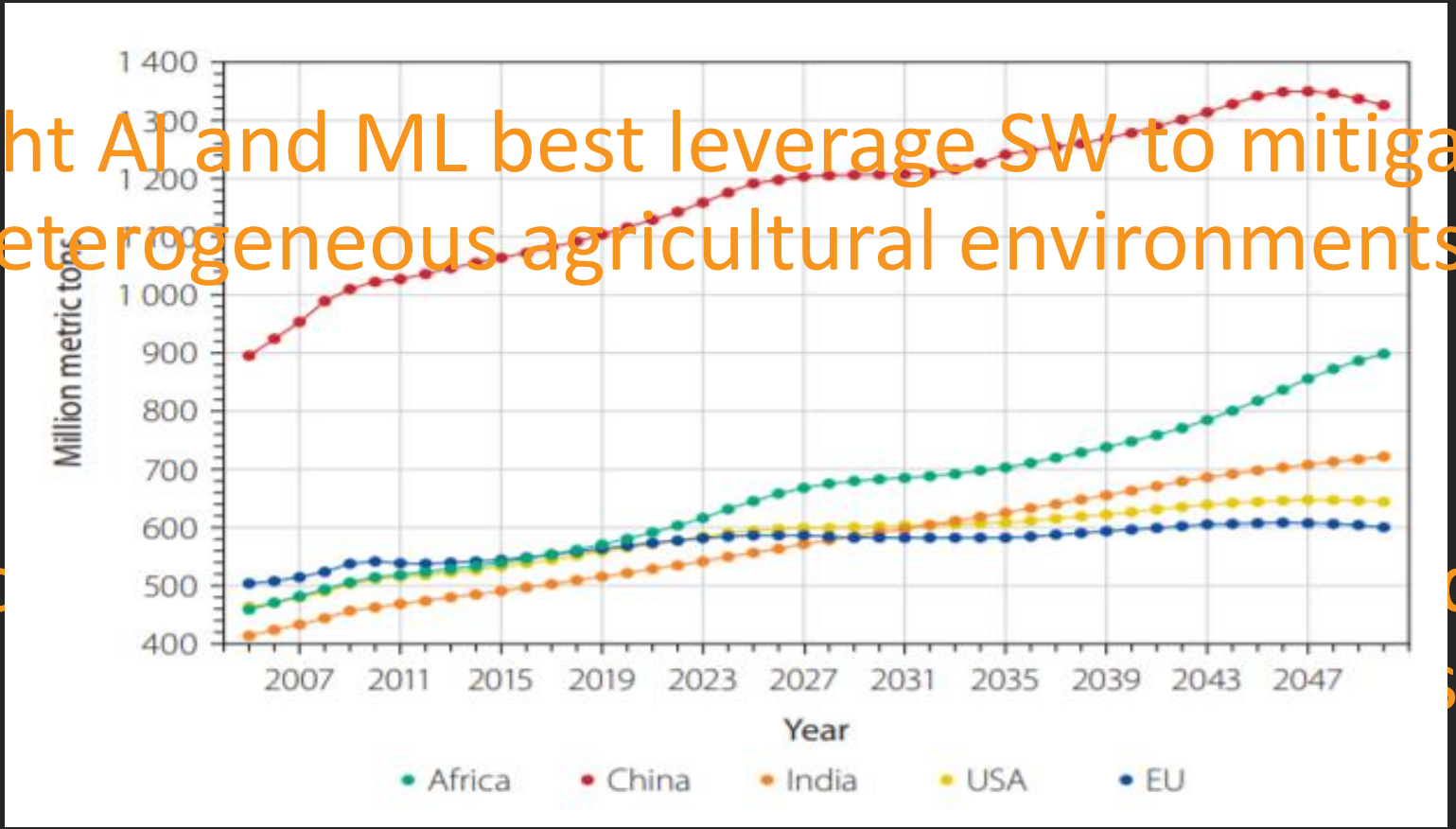
Artificial Intelligence for a Sustainable and Healthy Food System

1. SW is not AI and AI is not SW
2. AI is a field; SW is a project
3. SW owes a debt: used much from AI
4. SW should be a great playground for AI
5. AI projects should use SW to interoperate

T. Berners-Lee. https://www.w3.org/2006/Talks/0718-aaai-tbl/Overview.html#(2)

How might AI and ML best leverage SW to mitigate risk in heterogeneous agricultural environments?

What role d data play in harnessing s exemplar?



Can disruptive tech address key limitations such as poor soil fertility, inadequate access to labor, markets, credit, and technology in Africa?

(Cilliers, Hughes, and Moyer, 2011; https://issafrica.org/research/monographs/african-futures-2050)

Thank you!

CGIAR

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Big Data
in Agriculture

bigdata.cgiar.org